# Lecture 8: Transient Analysis of DC circuits

Transient Analysis is carried out to understand the dynamic behaviour of circuits, when one or more energy elements are present, such as inductance, capacitance. Two important aspects of transient analysis are:

1. It is often required to know how long it takes to reach s the circuit to respond to the applied source.
2. If there are two or more energy storing elements, they may interact to cause resonance phenomenon, which gives rise to high amplitude in voltage and current and can ruin the operation of the circuit or device functioning.

For most of DC circuits, the transient analysis can be carried out by solving differential equations. For most of circuits, it simplifies to to solving first order or second order differential equations of the nature, i.e.

First order linear diff. eqn :

$$\frac{di}{dt} + ai = b$$

$$\frac{dv}{dt} + av = b$$

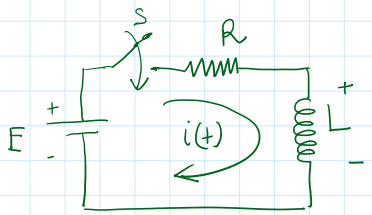
2nd order linear differential equation

$$\frac{d^2i}{dt^2} + a\frac{di}{dt} + bi = c$$

$$\frac{d^2v}{dt^2} + a\frac{dv}{dt} + bv = c$$

For example consider the following circuit.

Switch S is closed at $t=0$ (Assume no initial current in the inductance)




At $t = 0^-, 0, 0^+$

$$-E + Ri(t) + L\frac{di}{dt} = 0$$

$$\frac{di}{dt} + \frac{R}{L}i(t) = \frac{E}{L}$$

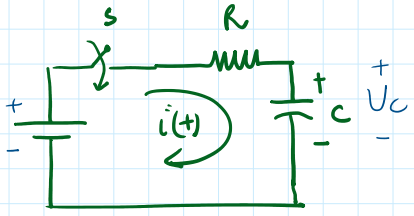
$$\frac{di}{dt} + a\,i(t) = b$$

$$a = \frac{R}{L}, \quad b = \frac{E}{L}$$

$$Ri(t) + v_c(t) = E$$

$$RC\frac{dv_c}{dt} + v_c(t) = E$$

$$\frac{dv_c}{dt} + \frac{1}{RC}v_c(t) = \frac{E}{RC}$$



$$\frac{dV_c}{dt} + \frac{1}{RC} V_c(t) = \frac{E}{RC}$$

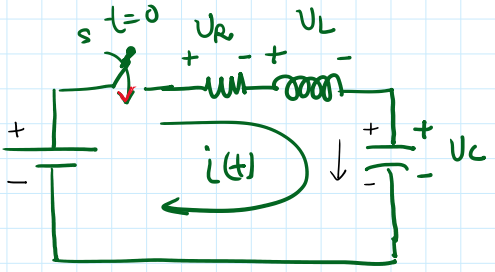
$$\frac{dV_c}{dt} + a\,V_c(t) = b$$

$$\Rightarrow a = \frac{1}{RC};\ b = \frac{E}{RC}$$

$$V_c = \frac{1}{C}\int i\,dt$$

$$i = C\frac{dV_c}{dt}$$

Now consider a second order system: A RLC Series Circuit




For $t \geq 0$

KVL: $V_R + V_L + V_C = E$

$$-E + V_R + V_L + V_C = 0$$

$$\downarrow$$

$$iR + L\frac{di}{dt} + \frac{1}{C}\int i\,dt = E \qquad \frac{de}{dt} \neq 0$$

$$e = E \Rightarrow \frac{dE}{dt} = 0$$

$$R\frac{di}{dt} + L\frac{d^2i}{dt^2} + \frac{i}{C} = 0$$

$$\frac{d^2i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{i}{LC} = 0 \Rightarrow \frac{d^2i}{dt^2} + a\frac{di}{dt} + bi = 0$$

where $a = \frac{R}{L}$, $b = \frac{1}{LC}$, $c = 0$

The above equation can also be arranged in $V_c$ and its derivative like this:

$$V_c = \frac{1}{C}\int i\,dt\ ; \quad i = C\frac{dV_c}{dt}\ ; \quad \frac{di}{dt} = C\frac{d^2V_c}{dt^2}$$

$$\Rightarrow \quad RC\frac{dV_c}{dt} + LC\frac{d^2V_c}{dt^2} + V_c = E$$

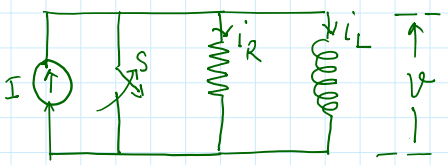
$$\Rightarrow \frac{d^2V_c}{dt^2} + \frac{R}{L}\frac{dV_c}{dt} + \frac{1}{LC}V_c = \frac{E}{LC}$$

which is of form $\frac{d^2V_c}{dt^2} + a\frac{dV_c}{dt} + bV_c = c$, where $a = \frac{R}{L}, b = \frac{1}{RC}, c = \frac{1}{LC}$

## Consider now take some examples of Parallel Circuits

(a) R-L Parallel Circuit:

Switch S is normally closed. It is opened at $t = 0$: using KCL




$$i_R + i_L = I$$

$$\frac{V_L}{R} + \frac{1}{L}\int V_L\,dt = I$$

$$\frac{1}{R}\frac{dV_L}{dt} + \frac{V_L}{L} = 0$$

$$v = V_L = L\frac{di_L}{dt}$$

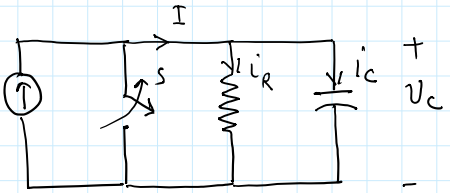
$$\Rightarrow i_L = \frac{1}{L}\int V_L\,dt$$

$$\frac{1}{R}\frac{dV_L}{dt} + \frac{V_L}{L} = 0$$

$$\Rightarrow \boxed{\frac{dV_L}{dt} + \frac{R}{L}V_L = 0}$$

$$\Rightarrow i_L = \frac{1}{L}\int V_L\,dt$$

Similarly: S is opened at $t=0$




$$i_R + i_C = I$$

$$\frac{V_C}{R} + C\frac{dV_C}{dt} = I$$

or $$\boxed{\frac{dV_C}{dt} + \frac{1}{RC}V_C = \frac{I}{C}}$$

Take now second order system: Parallel connected RLC system

$$I = i_R + i_L + i_C$$

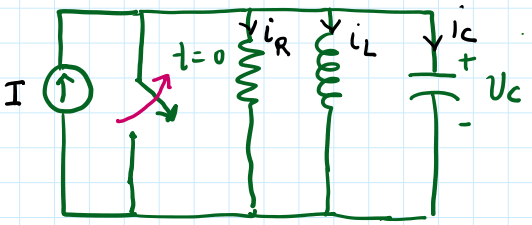
$$I = \frac{V_C}{R} + \frac{1}{L}\int V_C\,dt + C\frac{dV_C}{dt}$$

$$\downarrow$$

$$C\frac{d^2V_C}{dt^2} + \frac{1}{R}\frac{dV_C}{dt} + \frac{1}{L}V_C = 0$$

$$\frac{d^2V_C}{dt^2} + \frac{1}{RC}\frac{dV_C}{dt} + \frac{1}{LC}V_C = 0$$

$$\frac{d^2V_C}{dt} + a\frac{dV_C}{dt} + bV_C = c$$




$$L\frac{di_L}{dt} = V_C$$

$$i_L = \frac{1}{L}\int V_C\,dt$$

This can be arranged in current derivative also, i.e.

$$i_R + i_L + i_C = I$$

$$\frac{L}{R}\frac{di_L}{dt} + i_L + LC\frac{d^2i_L}{dt} = I$$

$$\Rightarrow \frac{d^2i_L}{dt} + \frac{1}{RC}\frac{di_L}{dt} + \frac{i_L}{LC} = \frac{I}{LC}$$

$$V_C = V_L = L\frac{di_L}{dt} = \frac{1}{C}\int i_C\,dt$$

$$\Rightarrow i_C = LC\frac{d^2i_L}{dt}$$

$$V_R = i_R R = L\frac{di_L}{dt}$$

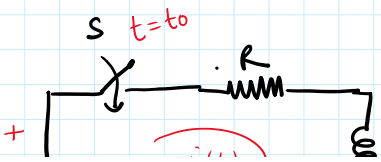
$$\Rightarrow i_R = \frac{L}{R}\frac{di_L}{dt}$$

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Now let us consider most common circuit we come across.

RL-Series Circuit:

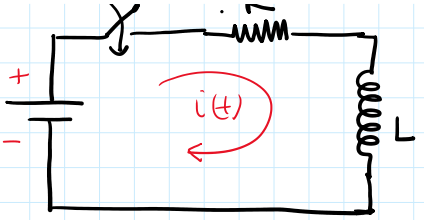
$$Ri + L\frac{di}{dt} = E$$



$$Ri + L\frac{di}{dt} = E$$

$$\frac{di}{dt} + \frac{R}{L} i = \frac{E}{L}$$

$$\frac{di(t)}{dt} + a\, i(t) = b$$




$$e^{at}\frac{di(t)}{dt} + e^{at} a\, i(t) = b e^{at}$$

$$\int \frac{d}{dt}\{e^{at} i(t)\} = \int b e^{at} dt$$

$$e^{at} i(t) = \int b e^{at} dt + k$$

$$e^{-at} e^{at} i(t) = e^{-at}\int b e^{at} dt + k e^{-at}$$

$$i(t) = k e^{-at} + e^{-at}\int b e^{at} dt$$

total response; transient response; steady state response

$$a = \frac{R}{L}; \quad b = \frac{E}{L}$$

$$i(t) = k e^{-\frac{R}{L}t} + e^{-\frac{R}{L}t}\int \frac{E}{L} e^{\frac{R}{L}t} dt$$

$$e^{-\frac{R}{L}t} \quad \frac{E}{L} \quad \frac{e^{\frac{R}{L}t}}{R/L}$$

$$i(t) = k e^{-\frac{R}{L}t} + \frac{E}{R}$$

$k e^{-\frac{R}{L}t}$: Transient solution; $\frac{E}{R}$: steady state solution

At $t = 0$, $i_L = i(0) = 0$

$$0 = k + \frac{E}{R}$$

$$K = -E/R; \quad i(t) = \frac{E}{R}\left(1 - e^{-\frac{R}{L}t}\right)$$

$$i(t) = \frac{E}{R}\left(1 - e^{-R/L\, t}\right)$$

$$i(t) = \frac{E}{R}\left(1 - e^{-t/\tau}\right)$$

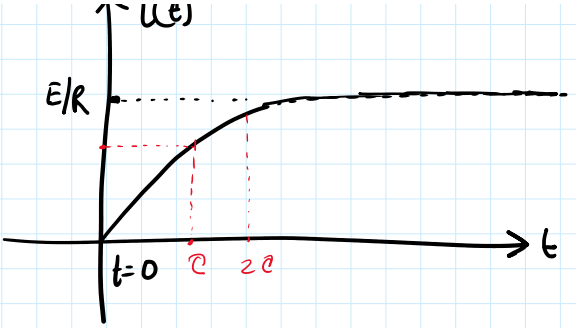
$\tau$ = Time constant of RL circuit = $L/R$ (Sec.)

When $t \to \infty$



When $t \to \infty$

$i_f = i_{steadystate} = E/R$




$t = \tau = \frac{L}{R}$

$i(\tau) = \frac{E}{R}\left(1 - e^{-1}\right)$

$e = 2.7182, \quad e^{-1} = 0.3679$

$i(\tau) = \frac{E}{R}\left(1 - 0.3679\right) = 0.6321 * \left(\frac{E}{R}\right)$

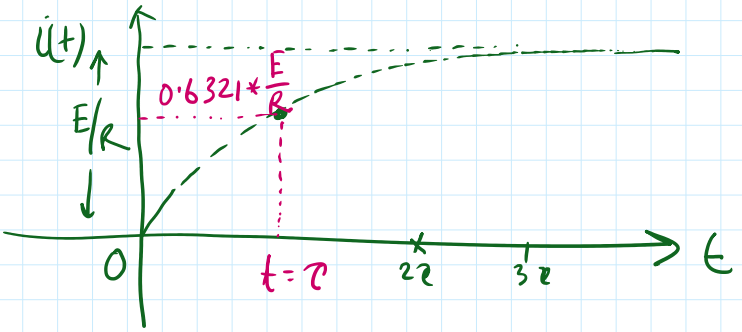
$= 63.21\%$ of final current

$i(2\tau) = \frac{E}{R}\left(1 - e^{-2}\right) = 0.86\ E/R$

$i(3\tau) = \frac{E}{R}\left(1 - e^{-3}\right) = 0.95\ E/R$

$i(4\tau) = \frac{E}{R}\left(1 - e^{-4}\right) = 0.98\ E/R$

$i(5\tau) = \frac{E}{R}\left(1 - e^{-5}\right) = 0.99\ E/R$




$i\left(t = t_0\right) = i_{L0}$

What is the effect of value of $\tau$ = Time constant of RL circuit?

High $\tau = \frac{L}{R}$ means poor time response, lethargic system

Low $\tau = \frac{L}{R}$ means fast time response – dynamic system.